Refined Project Plan

infrared.city-1: Tree detection

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1 Project Description

This chapter outlines the motivation, methods used, data sources, and the scope of this project.

1.1 Motivation and Project Overview

As urban areas expand and global temperatures rise due to climate change, being able to understand how trees can help cool cities has become increasingly important [1]. Furthermore, trees reduce the risk of flooding in cities and promote health and wellbeing [2]. Our project aims at taking the first step in the direction of better tree coverage in urban areas by creating a model that is able to predict tree locations using satellite imagery of densely populated areas.

Recently, there has been a rapidly growing body of research on urban tree detection using deep learning models. One of the most recent papers is by Gong et al. [3], who use satellite data not only to detect trees but also to estimate their size and density. For this, they use a slightly different model than the U-Net model used in this paper.

1.2 Data Sources

For our project, we use publicly available Sentinel-2 satellite images with 10-meter spatial resolution and four color channels, namely red, green, blue, and near-infrared. The images are retrieved via the API of the Copernicus Programme and are provided by the European Space Agency [4]. As seasonality has a great impact on the appearance of trees, imagery of different seasons will be used to train the model.

Additionally, we will use the publicly available dataset on every tree in the city of Vienna, the "Baumkataster", to train and test our model on exact tree locations [5].

1.3 Model

Previous papers suggest that for image segmentation with satellite data, the U-Net deep learning model is the most accurate for detecting objects, such as in street detection [6] as well as crop field detection [7]. Furthermore, last year's project also tried implementing a U-Net architecture but failed to yield promising results. This paper tries to build on that, especially because the data coaches also deem it a well-fitting model. This architecture will therefore also be used in this paper to predict tree locations.

The U-Net model is a special type of Convolutional Neural Network and consists of a contracting path and an expanding path. In the contracting path, convolution and pooling layers are used to extract features from the original imagery, which in this project will consist of four pictures for each season and four color channels each, as described above. The expanding path consists of the same structure of convolution and pooling layers, but used in the reverse direction to upsample the previously contracted imagery [8]. As suggested in similar work, this project will use skip connections between the expanding and contracting paths, as well as a temporal attention algorithm to prioritize more meaningful images [7]. The Binary Cross-Entropy loss between the predicted and actual locations will be used, and the actual locations from the “Baumkataster” will be split into training and test sets.

1.4 Project Scope and Possible Limitations

The model in this paper is trained only on satellite images of urban areas and is therefore only supposed to be used for tree detection within city environments. Additionally, the model focuses on tree detection in the temperate climate zone, as this is the vegetation it is trained to observe. Further analyses on the impact of trees on, for example, heat regulation in cities and flooding prevention are out of the scope of this project and could be the topic of future research.

Since this paper uses Sentinel-2 satellite images with 10-meter spatial resolution, issues could arise, as the images might not have a high enough resolution to detect trees reliably. Future work could therefore also focus on improving image quality by using better available satellite data.

1.5 Further Reading

The following additional papers are planned to be used as sources:

1. A review of individual tree crown detection and delineation from optical remote sensing images [9]
2. Integrating Traditional and Deep Learning Methods to Detect Tree Crowns in Satellite Images [10]
3. A Sentinel-2 machine learning dataset for tree species classification in Germany [11]
4. Object-based classification of urban plant species from very high-resolution satellite imagery [12]
5. Improving Urban Tree Species Classification with High Resolution Satellite Imagery and Machine Learning [13]

2 BEAM

Figure 1 presents the BEAM representation of the proposed tree detection system, illustrating the data sources, training and inference processes, model architecture, evaluation pipeline, and associated risks.

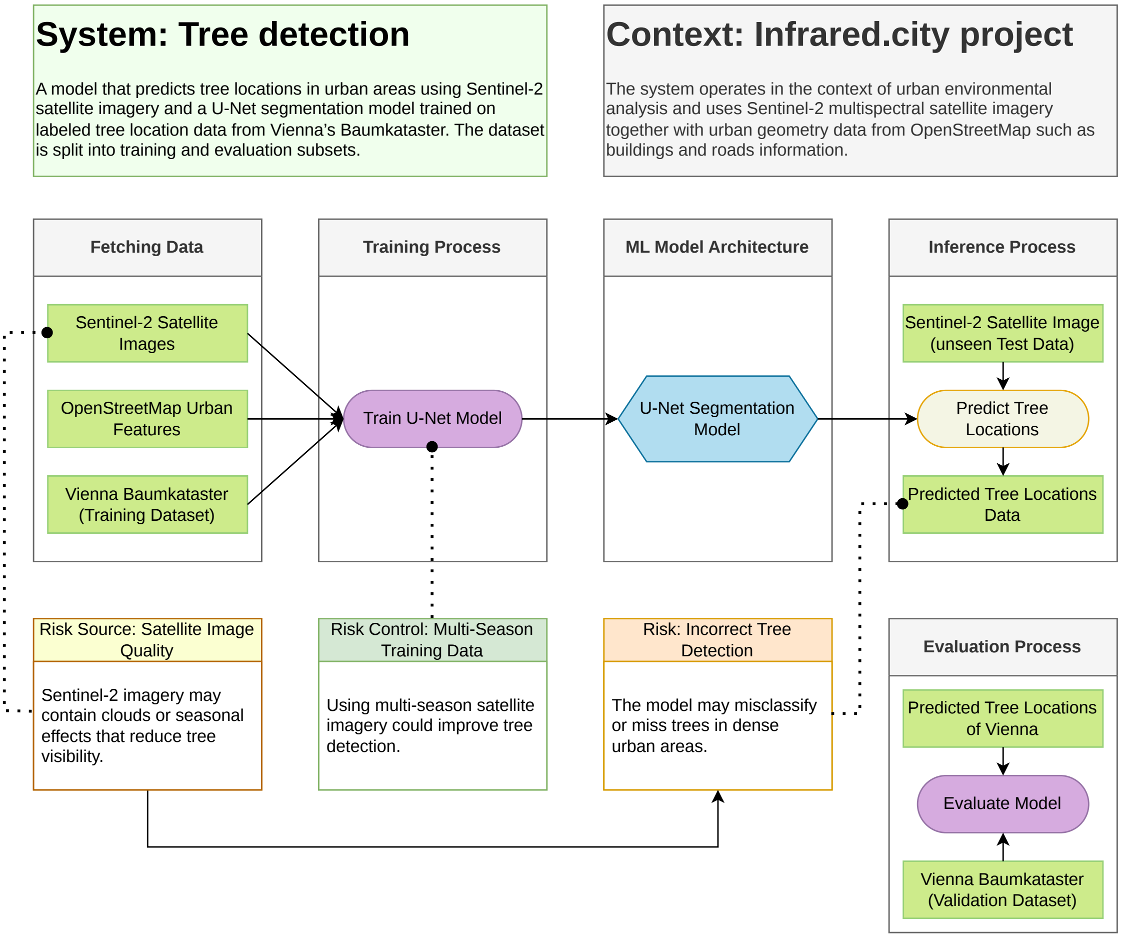

Figure 1: BEAM model of the proposed tree detection system.

3 Project Organisation

This chapter describes how the project is organised.

3.1 Team Roles

The main project team consists of 4 people, who are responsible for different focus points. All members commit to the project, and in case one person gets stuck, the focus is to work together and find a way to tackle the issue.

Person	Role	Tasks
Dominik Oberhumer	Organisation of Team	Preparation of Tools, Reminding of Deliverables and Timeplan
Moritz Hörmansdorfer	Data Scientist	Machine Learning Training of Model
Nina Salnikow	Data Engineer	Fetching and Preparation of Data
Sara Klasová	Data Scientist	EDA (Exploratory Data Analysis)

Besides the core team members, we also have additional support from people related to the project:

1. **Kavita Surana**: is our supervisor from WU. She will help us regarding the infrastructure and provides the necessary points for grading at the university.

2. **Oana Taut** and **Vasiliki Fragkia** are our data coaches from infrared.city. They will support us with different approaches and sources on how we can use the data to achieve our goal.

3.2 Communication outline

This section describes the communication styles we are committed to using throughout the project.

3.2.1 Written communication

We will use two Teams in Microsoft Teams for our basic communication.

1. **DsLab26S_infrared.city_1**: This Team contains two channels for two different approaches. The **General** channel is used as the communication channel with Kavita Surana (our connection to the lecture and the WU). The other channel (**team intern**) is accessible only by the team itself and is our main chat for communication throughout the project.

2. **DsLab_infrared.city_1-coaches**: This Team is the basis for communication with the data coaches from infrared.city. We were asked to separate this from the main communication source but still keep a Microsoft Teams Team for it. Thus, we created a different Team.

3.2.2 Regular Meetings and Communication

Our meetings will be held via Microsoft Teams in most cases. It is possible that we will be able to meet in person for specific meetings. However, we plan to keep the communication primarily remote or at least hybrid. We have already organized two regular weekly meetings for basic communication.

1. **Regular Team Meeting**: On Thursdays at 19:00, we will meet as a team remotely and discuss how to continue the work over the coming week.
2. **Irregular Meetings with Kavita Surana**: In cases where we need feedback from the WU side, we will schedule a meeting with her. We will schedule at least one meeting before the major deliverables. To easily schedule these meetings, we have access to a booking site for her time.
3. **Communication with Data Coaches from infrared.city**: No regular meetings are scheduled with the data coaches. Instead, we will write weekly reports in the shared Teams channel. In most cases, this weekly report will be sent by Dominik after our regular weekly team meeting.

3.2.3 Documentation of Meetings

Because of the many meetings, it is possible that someone may not be able to participate every time. To ensure communication with absent team members, we will write meeting notes and document the meeting output after every session. This documentation is mainly for us to better remember previous decisions. The protocols will be committed to our Git repository, like every other written result, as simple text documents containing the date and time, attendees, discussed topics, and follow-up decisions.

3.2.4 File and Code Repository

Every written result (documentation and project related) will be added to an already created GitHub repository. The repository is accessible by the team.[1]

3.2.5 Ticket and Issue Tracking

We have a Project in GitHub to track our progress on an issue and ticket level. This allows us to specify who is doing what and when, and it also creates an easy graphical overview.[2]

[1]See `https://github.com/nikmaster/wu_dslab_infrared-city`

[2]See `https://github.com/users/nikmaster/projects/2`

4 Timeplan

This chapter describes our planned timeline.

4.1 Milestones

The project will be conducted between March and June and is structured into several phases corresponding to the course milestones and deliverables.

4.1.1 Phase 1: Project Setup & Planning
March 9 – March 19

This initial phase focuses on defining the project scope, exploring available data sources, and establishing a solid project plan.

Tasks:

- Kick-off session and first meeting with data coaches (March 9)
- Define project objective and research question
- Explore available datasets (Sentinel-2, OpenStreetMap, etc.)
- Decide on modelling approach
- Assign team roles
- Draft project plan

Deliverable: Project Plan – March 19

4.1.2 Phase 2: Methodology Development
March 20 – March 29

During this phase, the primary goal is to acquire the necessary data and establish the foundational methodology for feature extraction.

Tasks:

- Data collection and preprocessing
- Feature extraction (spectral indices, vegetation signals)
- Define modelling pipeline
- Improve project plan based on feedback

Deliverable: Refined Project Plan – March 29

4.1.3 Phase 3: Data Processing & Initial Modelling
March 30 – April 21

This period is dedicated to executing the data processing pipeline and training the first iteration of machine learning models.

Tasks:

- Implement data processing pipeline
- Train initial machine learning models
- Evaluate early results
- Prepare intermediate analysis
- Create presentation slides

Deliverable: Intermediate Report and Slides – April 21

4.1.4 Phase 4: Model Improvement & Validation
April 22 – June 6

Focus now shifts towards optimizing model performance through rigorous validation, error analysis, and peer feedback.

Tasks:

- Intermediate consultations (April 22–24)
- Improve model performance
- Perform validation and error analysis
- Conduct sparring group meetings
- Prepare final report draft

Deliverable: Draft Final Report & Slides – June 7

4.1.5 Phase 5: Finalization & Presentation
June 8 – June 28

The final phase encompasses the conclusive evaluation of the models and the preparation of the final deliverables and class presentations.

Tasks:

- Final model evaluation
- Prepare final presentation
- Final presentation in class (June 18–19)
- Final report writing and revisions
- Submit sparring group minutes

Deliverables:

- Final Report – June 28
- Sparring Group Minutes – June 28

4.2 Tasks

Table 1 shows the current export of our issues from our issue tracker.
The export was done on March 28, 2026.

These tickets represent our current plan regarding who will do what and when. The table is structured by the major milestone phases.
In some cases, we do not have one specific assignee for a task. In these cases, we have currently assigned multiple people to a specific task, and we will decide later how to split up the work among them. For example, the writing of the report can be structured as subtasks by topic, with everybody writing different chapters. Since we are currently unsure about the exact topics, we decided that for now, it is sufficient to use multiple assignees.

Table 1: Project Timeline and Task Assignments

Task	Assignee(s)	Status	Start	End
Project Plan				
Setup Git Repository	Dominik Oberhumer	Done	10.03.2026	13.03.2026
Kickoff Session	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Done	13.03.2026	13.03.2026
Initialize Latex Template	Dominik Oberhumer	Done	13.03.2026	14.03.2026
Explore Datasets	Nina Salnikow	Done	13.03.2026	15.03.2026
Explore Modelling Approaches	Moritz Hörmansdorfer, Sara Klasová	Done	13.03.2026	16.03.2026
Write Project Plan Project Description	Moritz Hörmansdorfer	Done	14.03.2026	18.03.2026
Write Project Plan BEAM	Nina Salnikow	Done	14.03.2026	18.03.2026
Write Project Plan Timeplan	Sara Klasová	Done	14.03.2026	18.03.2026
Write Project Plan Communication	Dominik Oberhumer	Done	14.03.2026	18.03.2026
Write Project Plan Team Roles	Dominik Oberhumer	Done	14.03.2026	18.03.2026
Write Project Plan Literature	Sara Klasová	Done	14.03.2026	18.03.2026
Refined Project Plan				
Refine Project Plan Project Description	Moritz Hörmansdorfer	In Prog.	19.03.2026	29.03.2026
Refine Project Plan BEAM	Nina Salnikow	In Prog.	19.03.2026	29.03.2026
Refine Project Plan Timeplan	Dominik Oberhumer	In Prog.	19.03.2026	29.03.2026
Refine Project Plan Communication	Dominik Oberhumer	In Prog.	19.03.2026	29.03.2026
Refine Project Plan Team Roles	Dominik Oberhumer	In Prog.	19.03.2026	29.03.2026
Refine Project Plan Literature	Sara Klasová	In Prog.	19.03.2026	29.03.2026
Collect Data Baumkataster	Nina Salnikow	Done	19.03.2026	28.03.2026
Collect Data from Copernicus	Nina Salnikow	Todo	19.03.2026	28.03.2026

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Task	Assignee(s)	Status	Start	End
Collect Data additional Data sources?	Nina Salnikow	In Prog.	19.03.2026	28.03.2026
Define modelling pipeline	Moritz Hörmansdorfer, Nina Salnikow, Sara Klasová	Todo	19.03.2026	02.04.2026
Prepare Data Baumkataster (Features)	Nina Salnikow	Todo	29.03.2026	02.04.2026
Prepare Data Copernicus (Features)	Nina Salnikow	Todo	29.03.2026	02.04.2026
Prepare Data additional Data Sets?	Nina Salnikow	Todo	29.03.2026	02.04.2026
Intermediate Report				
Schedule minutes with sparing team	Dominik Oberhumer	Todo	30.03.2026	16.04.2026
Define structure of report	Dominik Oberhumer	Todo	30.03.2026	04.04.2026
EDA	Sara Klasová	Todo	03.04.2026	09.04.2026
Implement data processing pipeline	Moritz Hörmansdorfer, Nina Salnikow, Sara Klasová	Todo	03.04.2026	09.04.2026
Create latex template for report	Dominik Oberhumer	Todo	05.04.2026	08.04.2026
Write intermediate project report	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Todo	09.04.2026	21.04.2026
Prepare Slides for presentation	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Todo	09.04.2026	21.04.2026
Train initial machine learning models	Moritz Hörmansdorfer	Todo	10.04.2026	17.04.2026
Evaluate early results	Moritz Hörmansdorfer	Todo	10.04.2026	17.04.2026
Prepare intermediate analysis	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Todo	12.04.2026	17.04.2026
Draft Final Report & Slides				
Prepare latex for Draft Final Report	Dominik Oberhumer	Todo	21.04.2026	25.04.2026
Improve model performance	Moritz Hörmansdorfer, Sara Klasová	Todo	22.04.2026	21.05.2026
Define Resulted Database Structure	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Todo	22.04.2026	03.05.2026

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Task	Assignee(s)	Status	Start	End
Sparring group meetings	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Todo	22.04.2026	05.06.2026
Write Draft Final Report	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Todo	26.04.2026	07.06.2026
Perform validation and error analysis	Sara Klasová	Todo	28.04.2026	31.05.2026
Invite data coaches to presentation	Dominik Oberhumer	Todo	03.05.2026	09.05.2026
Implement Resulted Database	Moritz Hörmansdorfer, Sara Klasová	Todo	04.05.2026	31.05.2026
Prepare Slides for final presentation	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Todo	10.05.2026	07.06.2026
Final Report				
Prepare latex template for final report	Dominik Oberhumer	Todo	06.06.2026	09.06.2026
Prepare official sparring group minutes	Nina Salnikow	Todo	07.06.2026	28.06.2026
Final model evaluation	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Todo	08.06.2026	21.06.2026
Write final reports	Moritz Hörmansdorfer, Dominik Oberhumer, Nina Salnikow, Sara Klasová	Todo	09.06.2026	28.06.2026

4.3 GANTT Chart

Figure 2 provides a GANTT chart showing the structure of the tasks.

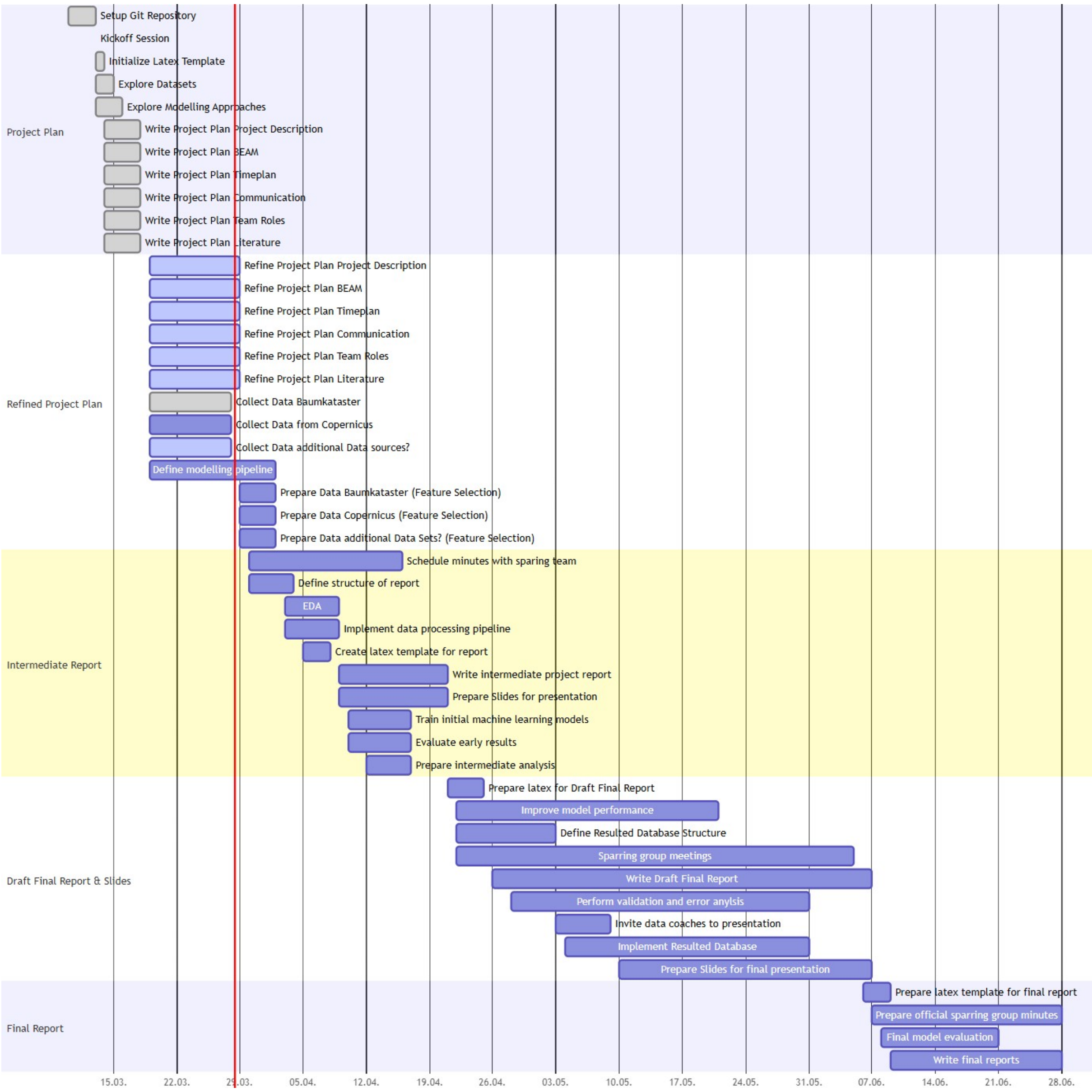

Figure 2: GANTT Diagramm

References

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